Failure of Metals

Generally, metals fail due to cracks initiated (1) in the metal from manufacture of operation, which continues to grow (2), and soon propagate (3) throughout the metal, which may cause catastrophic results.

There are 4 main types of metal failures:

Ductile Fracture

Brittle Fracture

Fatigue Failure

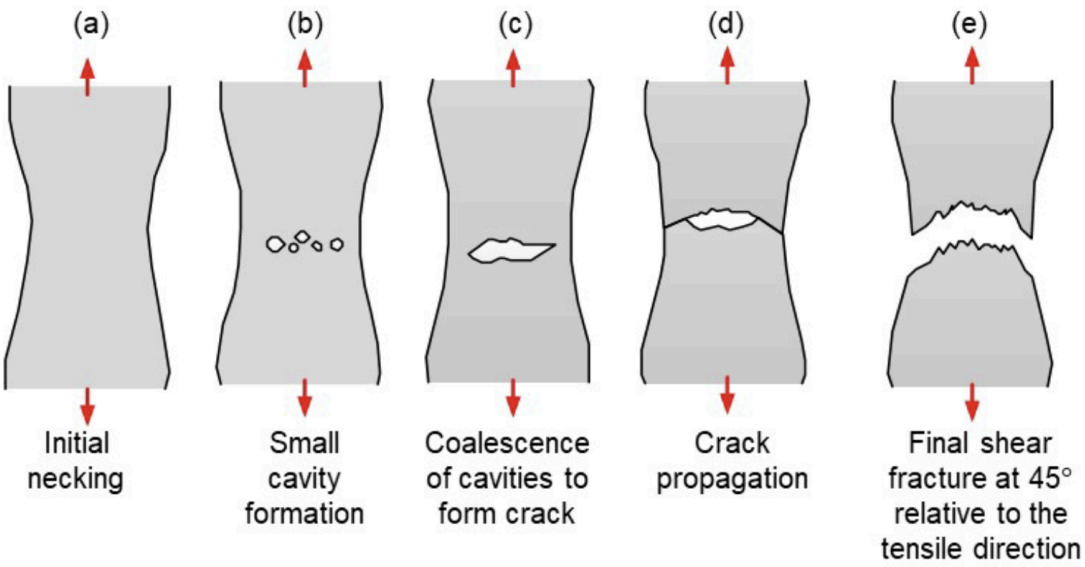
Creep

Ductile Fracture

Ductile fracture generally occurs when the structure has tensile stresses greater than the yield stress. This type of fracture is also known as an overload failure.

After the fracture, the metal has large amounts of deformation, with necking and cup & cone deformations at the fracture point.

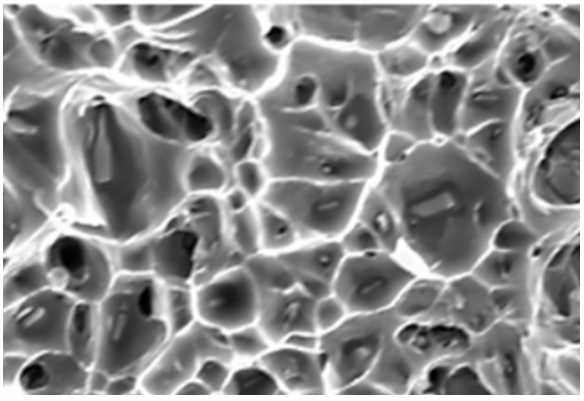
Fracture formation



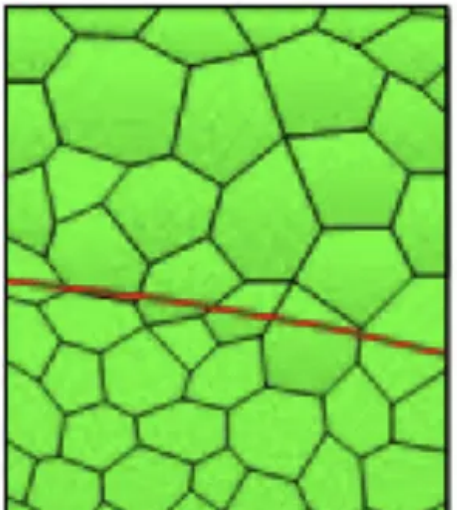
After necking, further tensile stresses are applied, causing microvoids, or cavities, to nucleate form at the neck. These cavities start to grow coalesce and join to form cracks. These cracks continue to propagate as more stress is applied, causing shear fractures in the metal 45 deg to the tensile direction.

At a macroscopic scale, the metal has a large deformation after the fracture, and due to the diagonal shear fractures, forms a cup-cone appearance at the ends. Like ductile materials after an impact test, the metal has a fibrous, dull and rough cross section.

Under a scanning electron microscope (SEM), dimples can be seen,



and under an optical microscope, transgranular cracks can be found across the grains.



Some causes of ductile failure could be that

Material is not strong enough, due to incorrect/improper heat treatment process, or improper material used.

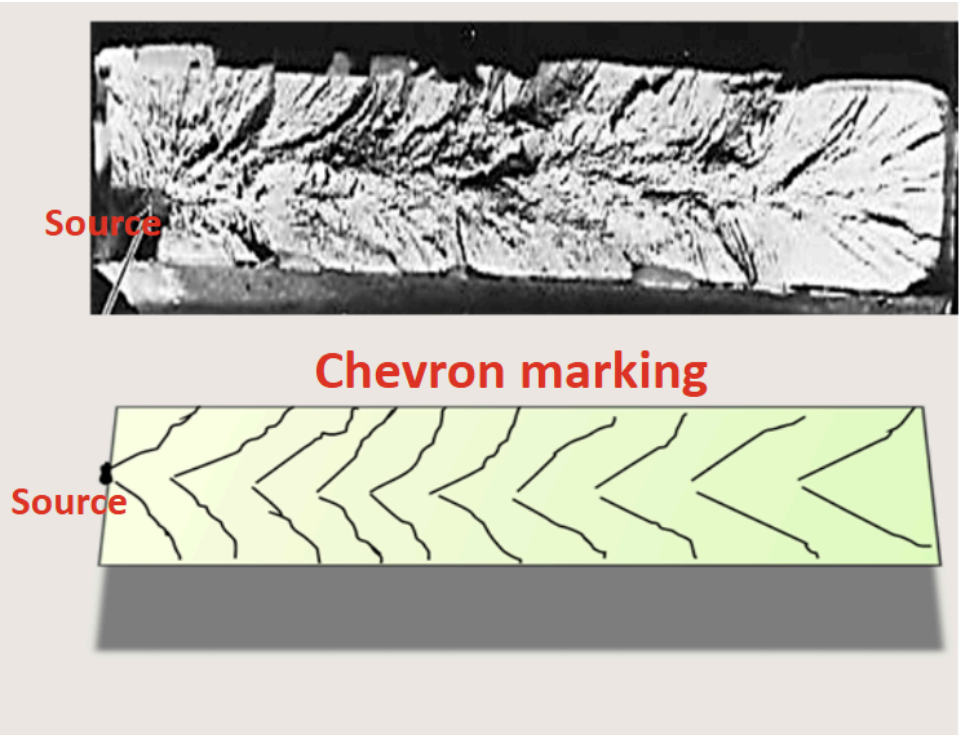
the service condition, or loads applied, were wrongly anticipated.

abnormal loading conditions were applied to the metal.

Brittle Failure

Brittle failure occurs when a small amount of work is absorbed during the failure, causing little deformation. Hence, it may occur at low tensile forces. It is also sudden, and its cracks propagate rapidly without warning, making is the most dangerous

At the fracture occurs in the elastic range, the metal has little to no deformations at the fracture, with a flat, shiny and crystalline at the fracture surface, with chevron, or river-like, markings.



Chevron markings are also directional, pointing to the source of the fracture, showing where the highest stress concentration was on the surface, and how it propagates through the metal.

Under an SEM, transgranular cracks across the grains or intergranular cracks (cracks that follow the grain boundaries) are seen as cleavages, and due to the fast propagation of cracks, causes a grainy and faceted texture.

Brittle failure is usually caused by three simultaneous conditions:

tensile stress is applied to the material, either due to external loading, or internal stresses, or both.

there are notches in the metal, either from a design feature, a pre-existing crack, or a sharp defect. As these notches disrupt the uniform stress throughout the material, the stresses concentrate and amplify around the notch, encouraging fracture.

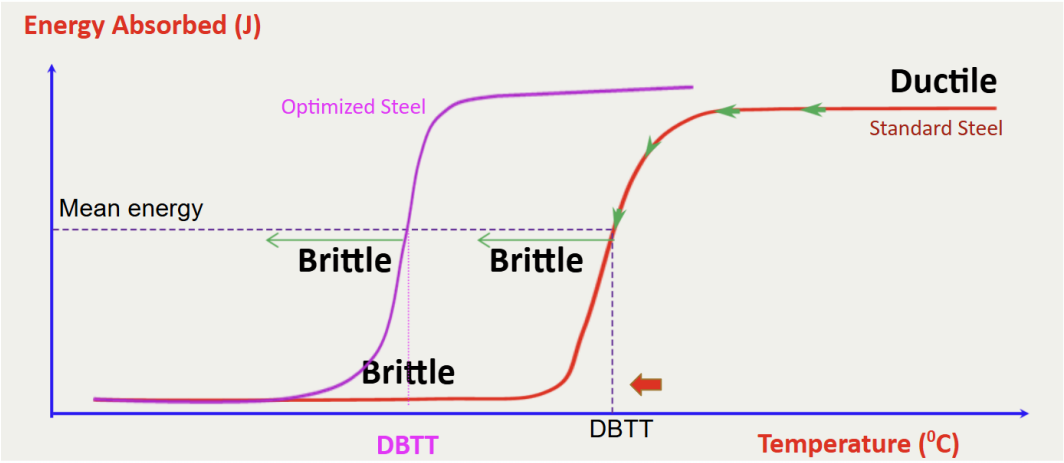
the ambient temperature is below the transition temperature, or the ductile-brittle transition temperature (DBTT)

DBTT is a range of temperatures where the fracture mode rapidly changes from ductile to brittle.

The value is determined by conducting multiple Charpy V notch tests at different temperatures, and plotting the energy absorbed at each temperature.

From the plot, the DBTT is then determined as the mean energy, or the midpoint of the transition between the energy absorbed

Hence, if the metal has a temperature below the DBTT.



For different materials, like comparing standard and optimised steel, optimised steel has a lower DBTT, meaning that as the temperature drops, standard steel becomes brittle earlier and loses its ability to absorb shock compared to optimised steel.

The DBTT also depends on the size and thickness of the specimen, rate of loading, and the metal's microstructure.

However, FCC metals like aluminium do not have a distinct DBTT, and should

To prevent brittle fracture, we can remove any of the three conditions causing brittle fracture, aka by

- removing any notches present on the metal
- removing the presence of tensile loads.
- making the ambient temperature greater than the DBTT.
- lowering the DBTT, via
  - decreasing the carbon content <0.15%
  - decreasing loading or impact rate
  - decreasing the notch depth, or increasing the notch radius (making the notch less sharp), to reduce stress concentration
  - refining the microstructure, by increasing nickel content (to 2-5%), or reducing grain size by adding grain refining elements, like Al and Nb.

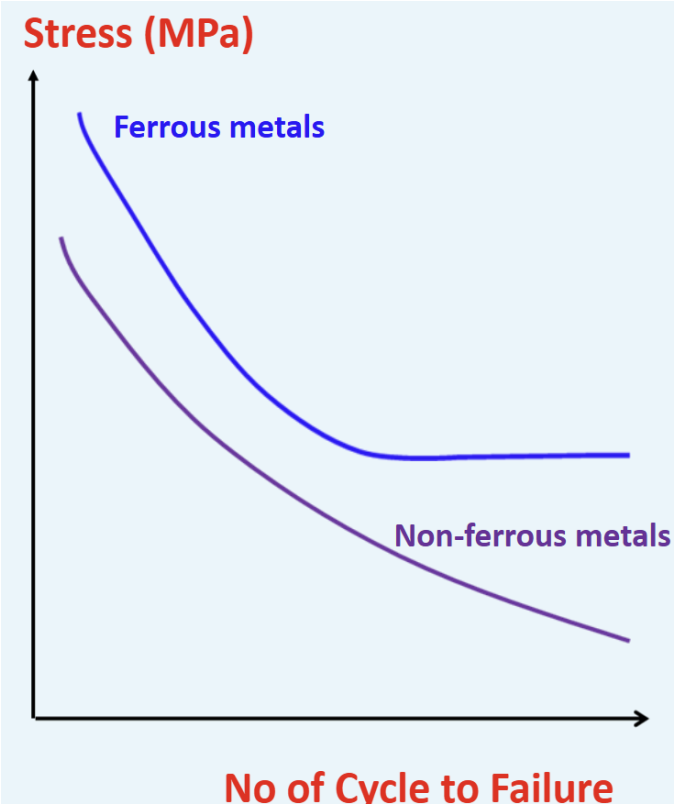
Fatigue Failure

Recall that fatigue describes failure under repetitive or cyclic loading. The stresses the material experiences is dynamic, i.e. the stresses change with time.

With enough cyclic or repetitive loading, the stress required for failure can be much lesser than the stress needed for failure under one instance of loading. This includes metals failing even when the maximum stresses experienced is below the yield stress, due to the numerous loading cycles, accounting for ~90% of all mechanical failures.

We can analyse how fatigue affects different metals with a fatigue test, where specimens are subjected to several repeated stresses until fracture occurs, or 10 million cycles have passed.

By recording the stress needed to cause failure after a certain number of cycles, we can plot the following, forming an S/N curve.



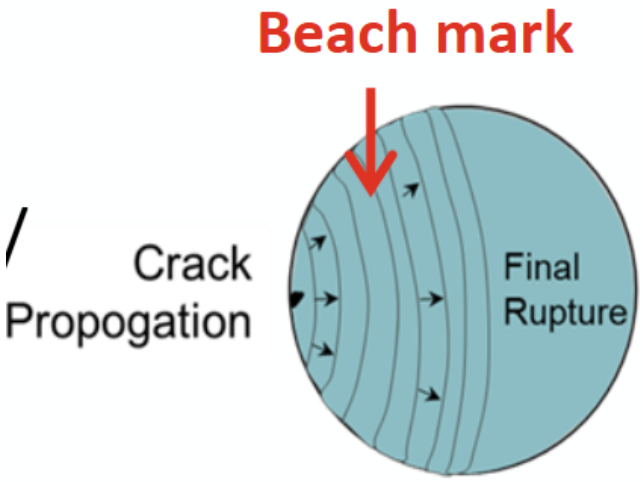
Generally, ferrous materials like steels have the stress flatten out at the endurance limit.

This means that if the specimen is stressed below the endurance limit, it will not undergo fatigue failure.

For non-ferrous metals, like aluminium, there is no limiting stress. Instead, the endurance strength at a certain number of cycles, e.g 10^6 cycles, is the stress where failure fatigue will not occur (i.e. below the S/N curve

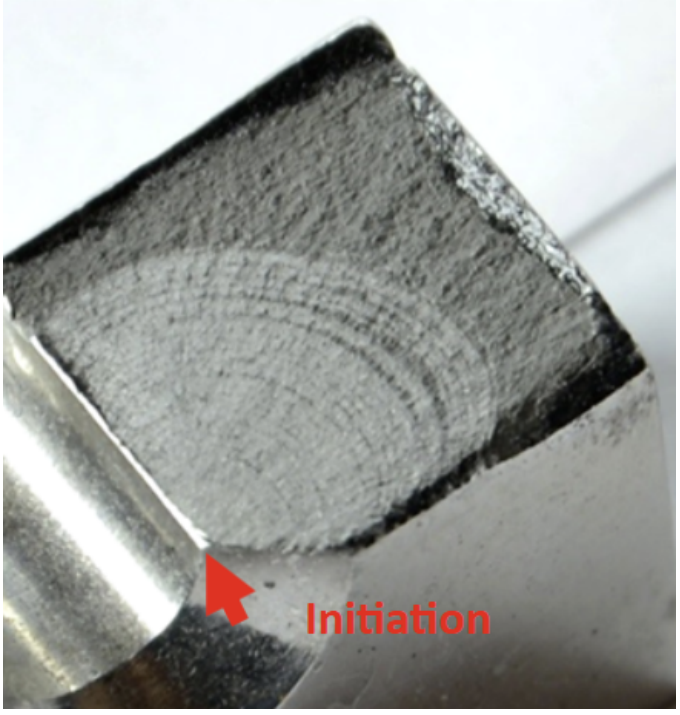
At a macroscopic scale, fatigue fracture has little to no deformation.

Fatigue fracture usually initiates from a point of highest stress concentration, usually from a crack or a narrow opening, propagating slowly throughout the material until significant, then rupturing and fracturing suddenly.



The crack propagation can be seen as a smooth section with beach marks/ripples on the surface of the fracture, as stress is dynamically applied with cycles.

The final rupture can be either crystalline or fibrous, depending if the final rupture is a ductile or brittle fracture, or both.



To prevent fatigue failure, we can reduce the stress concentration by

improving the design

Features like holes, notches, keyways and screw threads can cause stress concentration at these irregular geometries. Hence, these features should be minimised to reduce stress concentration and the risk of fatigue failure.

Other features like rounded contours, generous filets, chamfers, or having uniform sections (tapered shaft vs stepped shaft) can also reduce stress concentration.

improving the processing method

Certain processing methods like casting and powder processing can cause cracks, defects, voids, and inclusions, which can increase stress concentration at these locations.

Hence, efforts have to be made to reduce these side effects due to the processing.

Other methods like polishing and grinding may be used to ensure no cracks or irregularities are on the surface for stress concentrations to form.

improving the surface condition

Other than polishing the surface, we can also harden the surface to prevent cracks or irregularities from forming in the first place, via shot peening, cold rolling, and case hardening. These methods introduce compressive stresses to the surface to counter the tensile stresses.

Shot peening distorts the grains at the surface by using a stream of steel shots on the surface.

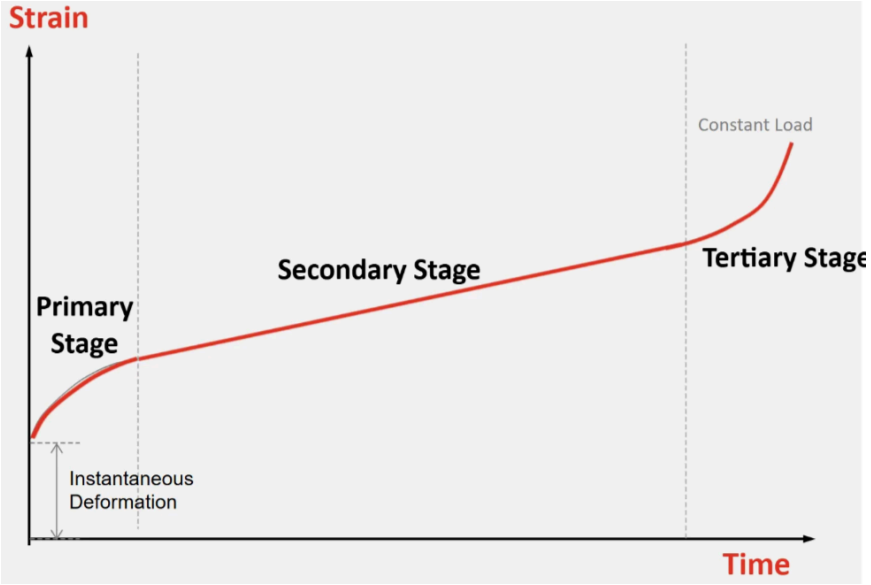
Case hardening involves using nitrogen or carbon to diffuse into the surface of the steel at elevated temperatures, forming a hard nitride or carbide surface layer.

Creep Failure

Creep is defined as a slow plastic deformation under a constant stress at elevated temperatures, typically below the yield strength of the metal.

Usually, creep occurs at temperatures greater than 1/3 of the melting point IN KELVIN!!!!!!!!!!!!!!!!!

There are three main stages of creep, shown by the creep curve, plotting the strain against time under a constant load.



Primary Stage

Deformation occurs at a fast rate at first, but slows down over time, due to work hardening

Secondary Stage

The specimen continues to deform at a constant strain rate, at a relatively low rate, as work hardening is slowly balanced out by thermal softening. The strain rate is also affected by the grain size, with finer grains creeping faster than larger grains.

Tertiary Stage

Strain rate accelerates as microstructure deteriorates, and as necking, voids and cracks form in the metal, until fracture occurs.

The creep curve is also temperature and load dependent, as a higher temperature per load causes it to strain more, causing the curve to look narrower and it fracturing faster.

To prevent creep, we can

Reduce the working temperature

using alloys of higher melting point.

reducing constant load supplied

increasing grain size (coarser grains) or single crystal alloys

strengthening the alloy

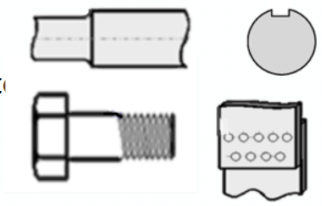
solid solution strengthening

Precipitation/dispersion hardening

Generally, the prevention methods are common as:

- Poor design

  Stress raisers such as sharp corners, keyways, holes et

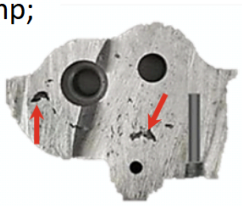


- Material selection

  Considering i) type of loading; ii) environment, temp;
  iii) rate of wear/erosion

- Imperfection in materials

  Surface defects, Internal flaws
  Reduce overall strength and act as stress raiser



- Deficiencies in processing

  Produce high residual stresses, cracks

- Improper service conditions

  Operating the machine at conditions beyond that specified by the manufacturer.

- Inadequate maintenance